University of Caloocan City
Computer Studies Department
CCS 119 – Capstone Project

EduPredict: Questionnaire Scoring, Computation, and Supporting Studies

Scoring and research basis (consultation draft)

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1. Purpose and Status

At our previous defense the panel asked for two things: how the questionnaire is scored and computed, and which studies support the questions and the results of EduPredict. This paper covers both. The 36 questionnaire items are in a separate file, EduPredict_Questionnaire.docx.

The questionnaire is still a draft for expert validation and has not been validated for CLAS students of the University of Caloocan City (UCC). The employability readiness model and the dropout model are not trained yet. We are waiting for the revised questionnaire and the final set of dropout features before training them.

2. The Instrument at a Glance

The questionnaire has 36 items in six sections, and every item uses a 5-point scale. Only two sections, Competence and Psychological Capital, are used as inputs to a machine learning model. The other sections are context. They help explain the results and support the recommendations, and the system shows them apart from the model predictions.

Code	Section / construct	Items	Role in EduPredict
A	Academic factors (academic self-perception)	6	Context and recommendations; compared with actual grades. Not an ML input.
S	Socioeconomic factors (financial and access context)	6	Context and support recommendations. Not an ML input.
B	Behavioral factors (student engagement)	6	Engagement summary and intervention rules. Does not change the dropout probability.
C	Competence / employability skills	8	ML input: Competence mean for the employability readiness model.

Code	Section / construct	Items	Role in EduPredict
P	Psychological capital / career readiness	6	ML input: PsyCap mean for the employability readiness model.
E	Self-perceived employability	4	Used only to compare with the model result. Not an input, to avoid target leakage.

Grades, skills, certifications, projects, and internship or work records come from other parts of the system, so the questionnaire does not ask for them again.

3. Scoring of the Questionnaire and Its Computation

3.1 Response scale and coding

Each item is answered from 1 (Strongly Disagree) to 5 (Strongly Agree), where 2 is Disagree, 3 is Neutral, and 4 is Agree. All 36 items are worded positively, so no item needs reverse scoring. We do not add up all 36 items, since each section measures a different thing. Every section is scored on its own.

3.2 Construct score (mean)

The score of a section is the mean of the student's answers in that section:

Construct mean = sum of the student's responses ÷ number of items in the section

Sections A, S, B, and P have 6 items, section C has 8, and section E has 4. The result is always between 1.00 and 5.00. We use the mean instead of the sum so that sections with different numbers of items can be compared. Describing a multi-item Likert scale with a mean is accepted practice (Boone & Boone, 2012).

3.3 Interpretation of scores

To get the verbal levels, we divided the 1–5 scale into five equal intervals, (5 − 1) ÷ 5 = 0.80 per level (Pimentel, 2010). The readiness percentages in the table use the conversion in Section 3.4 on the same boundaries.

Mean score	Verbal interpretation	Readiness % (employability)	Context flag (A, S, B)
1.00 – 1.80	Very Low	0 – 20	Flag
1.81 – 2.60	Low	above 20 – 40	Flag
2.61 – 3.40	Moderate	above 40 – 60	No flag
3.41 – 4.20	High	above 60 – 80	No flag
4.21 – 5.00	Very High	above 80 – 100	No flag

These bands are only a descriptive convention. They are not cut-offs that have been tested against data, and we list this as a limitation. The context flag, which is a mean of 2.60 or below in A, S, or B, is a rule we set in the system to trigger a recommendation. It is not a model prediction.

3.4 Employability Readiness Score

Only the Competence mean (C1–C8) and the Psychological Capital mean (P1–P6) are sent to the employability model. The steps are:

1. Compute CompetenceMean and PsyCapMean using the formula in Section 3.2.
2. The model returns a predicted employability score on its own scale, roughly 1 to 5.
3. Convert the score to a percentage for display:

$$Readiness\ \% = ((score - 1) \div 4) \times 100$$

The value shown to the student is called the Employability Readiness Score. It does not mean the student has that percent chance of being hired. Each prediction is saved with the raw model score, the displayed percentage, the model version, the dataset version, and the time, so every result can be traced.

3.5 Worked example

Construct	Responses	Sum	Items	Mean	Level
Competence (C1–C8)	4, 5, 4, 3, 4, 4, 5, 4	33	8	4.125	High
Psych. Capital (P1–P6)	4, 4, 5, 4, 3, 4	24	6	4.000	High
Self-perceived employability (E1–E4)	4, 4, 3, 4	15	4	3.750	High

Say the model returns 3.90 for the Competence and PsyCap means above. This number is made up for the example, since the model is not trained yet. Then Readiness % = ((3.90 − 1) ÷ 4) × 100 = 72.5%, which is in the High band. The student's own Self-Perceived Employability mean of 3.75 becomes ((3.75 − 1) ÷ 4) × 100 = 68.75% with the same formula. The two values differ by about 3.75 percentage points, so the student's self-view and the model result are close. If the gap were large, we would show it to the student and staff as something to talk about, and not as an error.

3.6 Use of the context sections

The Academic (A), Socioeconomic (S), and Engagement (B) means are read using the same table in Section 3.3. They are not inputs to either model. A low Socioeconomic mean, for example, can trigger a financial-support recommendation, and a low Engagement mean can trigger an action from the predefined intervention list. The dropout model only uses academic records and profile fields that can be mapped to the selected UCI predictors, and the Behavioral items B1–B6 do not change its probability.

4. Computation Used to Validate the Instrument and the Models

4.1 Content validity (expert validation)

Each expert rates every item on relevance, clarity, and appropriateness for UCC CLAS, using a 1–4 scale. The item-level content validity index is:

I-CVI = number of experts who rated the item 3 or 4 ÷ total number of experts

The scale-level index, S-CVI/Ave, is the average of all the I-CVIs. The usual benchmarks are an I-CVI of at least 0.78 and an S-CVI/Ave of at least 0.90 (Lynn, 1986; Polit & Beck, 2006). Items that fall below the benchmark will be revised or removed, and we will record every change.

4.2 Reliability (pilot test)

After the revision, we will pilot the questionnaire with students similar to the intended CLAS users. Internal consistency is computed for each construct separately instead of one value for all 36 items. We use Cronbach's alpha (Cronbach, 1951):

$$\alpha = (k \div (k - 1)) \times (1 - \Sigma \text{ item variances} \div \text{variance of the total score})$$

Here k is the number of items in the construct, which is 8 for Competence and 6 for Psychological Capital. A value of 0.70 or higher is generally accepted (Nunnally & Bernstein, 1994). We will look most closely at Competence and PsyCap because these two feed the model. We will not delete an item just to raise alpha, because the meaning of the construct has to be kept.

4.3 Employability model (regression)

The employability target is a continuous score, so the model is judged by error measures and not by accuracy:

$$MAE = (1 \div n)\ \Sigma\ |\text{actual} - \text{predicted}|$$

$$RMSE = \sqrt{[(1 \div n)\ \Sigma\ (\text{actual} - \text{predicted})^2]}$$

$$R^2 = 1 - (\text{residual sum of squares} \div \text{total sum of squares})$$

The external dataset has only 266 records. Because of this, we will start with simple models (Linear or Ridge Regression, Random Forest Regressor, and Gradient Boosting Regressor) and use repeated cross-validation instead of a single train/test split.

4.4 Dropout model (classification)

The dropout model will be evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and a confusion matrix. The Low, Moderate, and High risk thresholds will come from validation and calibration results, and we will not simply split the probability into thirds.

5. Supporting Studies for the Questions and the Results

5.1 Academic factors (A1–A6)

These items measure academic self-efficacy, meaning how confident a student is in handling coursework. Chemers, Hu, and Garcia (2001) found that academic self-efficacy was related to the performance and adjustment of first-year college students. The idea comes from Bandura's (1997) self-efficacy theory. Tinto's (1975) model of dropout is the classic basis for linking a student's academic experience to the decision to stay or leave. In EduPredict these answers are

only context. We compare them with the actual grades, and they do not replace academic records.

5.2 Socioeconomic factors (S1–S6)

Financial strain is a known source of stress for undergraduates, and Northern, O'Brien, and Goetz (2010) developed and validated a financial stress scale for it. Items S1, S5, and S6 (budget, focus on studies, and financial support) follow this idea. Items S2 to S4 (transportation, device, and internet) were written by us for the UCC setting, so we ask the adviser and the expert validators to check them carefully.

5.3 Behavioral factors / engagement (B1–B6)

Fredricks, Blumenfeld, and Paris (2004) describe student engagement as having behavioral, emotional, and cognitive parts. Maroco, Maroco, Campos, and Fredricks (2016) built the University Student Engagement Inventory (USEI) to measure these parts in university students. Our B items cover the behavioral part: attendance, study time, participation, use of learning materials, meeting deadlines, and persistence. The answers are used for the engagement summary and the intervention rules. They are not dropout model inputs.

5.4 Competence (C1–C8) and Psychological Capital (P1–P6)

These two constructs are the direct basis of the employability model. They come from the open dataset of Pianda, Hilmiana, Widianto, and Sartika (2026), which has survey data on internship experience, competence, psychological capital, and employability. According to the dataset record, the data were gathered by simple random sampling from August to December 2022 from 266 vocational students in Aceh Province, Indonesia, all of whom had done an industry internship. The same research group also wrote a review of the literature on internship experience and the employability of vocational students (Pianda et al., 2024), which supports their framework.

Psychological capital consists of hope, optimism, resilience, and self-efficacy (Luthans, Avolio, Avey, & Norman, 2007). For the results side, Xiong et al. (2021) surveyed about 6,700 higher vocational students and found that psychological capital was significantly correlated with employability. This supports using PsyCap to estimate employability readiness.

5.5 Self-perceived employability (E1–E4)

Rothwell, Herbert, and Rothwell (2008) built and validated a self-perceived employability scale for university students. Van der Heijde and Van der Heijden (2006) and Fugate, Kinicki, and

Ashforth (2004) treat employability as a mix of skills, adaptability, and personal factors. In EduPredict the E items are only used to compare the student's own view with the model result, and they are not model inputs.

5.6 Supporting the dropout results

The dropout model uses the "Predict Students' Dropout and Academic Success" dataset (Realinho, Machado, Baptista, & Martins, 2022). Its authors describe it as demographic, socioeconomic, macroeconomic, and academic data about enrollment and performance at the end of the first and second semesters, made for building models that predict dropout and academic success.

Kim, Yoo, and Kim (2023), in a preprint, used the same records, keeping only graduates and dropouts (3,630 students), and compared four classifiers. Random Forest did best with a ROC-AUC of 0.955. When all academic features were removed, the average ROC-AUC of the four classifiers dropped from 0.935 to 0.811, which was a larger drop than when demographic, socioeconomic, or macroeconomic data were removed. This is why we plan to feed the dropout model mostly with academic records. Moreira da Silva et al. (2022) used mostly student grades at a Portuguese university and reported an F1-score of 81% on the test set, so our target of 70–80% looks realistic. Alyahyan and Düştegör (2020) give a wider review of methods for predicting academic success in higher education.

6. Limitations

The employability dataset is not from UCC. It comes from vocational students in Indonesia, so the results need local validation before we can say they apply to UCC CLAS students. The dataset is also small (266 records), so we have to keep the model simple and our claims careful. The score is a modeled readiness construct based on self-reported answers. It is not a verified probability of getting a job.

The UCI dropout dataset comes from Portuguese higher education, and some of its variables and grade scales cannot be moved directly to UCC. The UCC grades on the 1.00–5.00 scale, for example, cannot just replace the Portuguese 0–20 scale, so we need a documented conversion. The engagement and socioeconomic items give useful context but are not model predictors unless we collect local outcome data later. Self-reported answers can also be affected by bias or inaccurate responses. In future work we should collect UCC outcomes on retention, dropout, and

graduate employment, with consent and privacy controls (RA 10173), so the models can be retrained and validated locally.

References

Alyahyan, E., & Düştegör, D. (2020). Predicting academic success in higher education: Literature review and best practices. *International Journal of Educational Technology in Higher Education, 17*, Article 3.

Bandura, A. (1997). *Self-efficacy: The exercise of control*. W. H. Freeman.

Boone, H. N., & Boone, D. A. (2012). Analyzing Likert data. *Journal of Extension, 50*(2).

Chemers, M. M., Hu, L., & Garcia, B. F. (2001). Academic self-efficacy and first-year college student performance and adjustment. *Journal of Educational Psychology, 93*(1), 55–64.

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika, 16*(3), 297–334.

Fredricks, J. A., Blumenfeld, P. C., & Paris, A. H. (2004). School engagement: Potential of the concept, state of the evidence. *Review of Educational Research, 74*(1), 59–109.

Fugate, M., Kinicki, A. J., & Ashforth, B. E. (2004). Employability: A psycho-social construct, its dimensions, and applications. *Journal of Vocational Behavior, 65*(1), 14–38.

Kim, S., Yoo, E., & Kim, S. (2023). *Why do students drop out? University dropout prediction and associated factor analysis using machine learning techniques* (arXiv:2310.10987) [Preprint]. arXiv. https://arxiv.org/abs/2310.10987

Luthans, F., Avolio, B. J., Avey, J. B., & Norman, S. M. (2007). Positive psychological capital: Measurement and relationship with performance and satisfaction. *Personnel Psychology, 60*(3), 541–572.

Lynn, M. R. (1986). Determination and quantification of content validity. *Nursing Research, 35*(6), 382–385.

Maroco, J., Maroco, A. L., Campos, J. A. D. B., & Fredricks, J. A. (2016). University student's engagement: Development of the University Student Engagement Inventory (USEI). *Psicologia: Reflexão e Crítica, 29*, Article 21.

Moreira da Silva, D. E., Solteiro Pires, E. J., Reis, A., Moura Oliveira, P. B., & Barroso, J. (2022). Forecasting students dropout: A UTAD University study. *Future Internet, 14*(3), 76. https://doi.org/10.3390/fi14030076

Northern, J. J., O'Brien, W. H., & Goetz, P. W. (2010). The development, evaluation, and validation of a financial stress scale for undergraduate students. *Journal of College Student Development, 51*(1), 79–92.

Nunnally, J. C., & Bernstein, I. H. (1994). *Psychometric theory* (3rd ed.). McGraw-Hill.

Pianda, D., Hilmiana, Widianto, S., & Sartika, D. (2024). The impact of internship experience on the employability of vocational students: A bibliometric and systematic review. *Cogent Business & Management, 11*(1), 2386465.

Pianda, D., Hilmiana, Widianto, S., & Sartika, D. (2026). A stratified survey dataset on internship experience, competencies, psychological capital, and employability of vocational students in Aceh, Indonesia. *Data in Brief, 67*, 112992. https://doi.org/10.1016/j.dib.2026.112992

Pianda, D., Hilmiana, Widianto, S., & Sartika, D. (2026). *Dataset of student internship experience, competence, psychological capital and employability's vocational student (Original data)* (Version 4) [Data set]. Mendeley Data. https://doi.org/10.17632/hmn3b4c6c4.4

Pimentel, J. L. (2010). A note on the usage of Likert scaling for research data analysis. *USM R&D Journal, 18*(2), 109–112.

Polit, D. F., & Beck, C. T. (2006). The content validity index: Are you sure you know what's being reported? Critique and recommendations. *Research in Nursing & Health, 29*(5), 489–497.

Realinho, V., Machado, J., Baptista, L., & Martins, M. V. (2022). Predicting student dropout and academic success. *Data, 7*(11), 146.

Realinho, V., Vieira Martins, M., Machado, J., & Baptista, L. (2021). *Predict students' dropout and academic success* [Data set]. UCI Machine Learning Repository. https://doi.org/10.24432/C5MC89

Rothwell, A., Herbert, I., & Rothwell, F. (2008). Self-perceived employability: Construction and initial validation of a scale for university students. *Journal of Vocational Behavior, 73*(1), 1–12.

Tinto, V. (1975). Dropout from higher education: A theoretical synthesis of recent research. *Review of Educational Research, 45*(1), 89–125.

Van der Heijde, C. M., & Van der Heijden, B. I. J. M. (2006). A competence-based and multidimensional operationalization and measurement of employability. *Human Resource Management, 45*(3), 449–476.

Xiong, L., Zhang, Q., Cao, Y., Yang, J., & Yang, M. (2021). An exploration into the influence of higher vocational students' psychological capital on their employability. *SHS Web of Conferences, 96*, 02007. https://doi.org/10.1051/shsconf/20219602007